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Learn > Passive Reconnaissance



Passive Reconnaissance

Learn about the essential tools for passive reconnaissance, such as whois, nslookup, and dig.

60 min 2,08,303

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Task 1 Introduction

Welcome to the first room of the Network Security Module. This module covers:

1. [Passive Reconnaissance](#)
2. [Active Reconnaissance](#)
3. [Nmap Live Host Discovery](#)
4. [Nmap Basic Port Scans](#)
5. [Nmap Advanced Port Scans](#)
6. [Nmap Post Port Scans](#)
7. [Protocols and Servers](#)
8. [Protocols and Servers 2](#)
9. [Network Security Challenge](#)

In this room, after we define passive reconnaissance and active reconnaissance, we focus on essential tools related to passive reconnaissance. We will learn three command-line tools:

- `whois` to query WHOIS servers
- `nslookup` to query DNS servers
- `dig` to query DNS servers

We use `whois` to query WHOIS records, while we use `nslookup` and `dig` to query DNS database records. These are all publicly available records and hence do not alert the target.

We will also learn the usage of two online services:

- DNSDumpster
- Shodan.io

These two online services allow us to collect information about our target without directly connecting to it.

Pre-requisites: This room requires basic networking knowledge along with basic familiarity with the command line. The modules [Network Fundamentals](#) and [Linux Fundamentals](#) provide the required knowledge if necessary.

Important Notice: Please note that if you're not subscribed, the AttackBox won't have Internet access, so you will need to use the VPN to complete the questions that require Internet access.

Answer the questions below

This room does not use a target virtual machine (VM) to demonstrate the discussed topics. Instead, we will query public WHOIS servers and DNS servers for domains owned by TryHackMe. Start the AttackBox and make sure it is ready. You will use the AttackBox to answer the questions in later tasks, especially tasks 3 and 4.

No answer needed

Correct Answer

Task 2 Passive Versus Active Recon

Learn > Passive Reconnaissance



Passive Reconnaissance

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📊 63 min ⌚ 2/06/2023 ⚡ 🔒

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Task 1 Introduction

Task 2 Passive Versus Active Recon

This room expects the user to have a working knowledge of computer networks. If you like to brush up on this topic, you are encouraged to study the [Network Fundamentals](#) module first.

Before the dawn of computer systems and networks, in the Art of War, Sun Tzu taught, "If you know the enemy and know yourself, your victory will not stand in doubt." If you are playing the role of an attacker, you need to gather information about your target systems. If you are playing the role of a defender, you need to know what your adversary will discover about your systems and networks.

Reconnaissance (recon) can be defined as a preliminary survey to gather information about a target. It is the first step in [The Unified X88 Chain](#) to gain an initial foothold on a system. We divide reconnaissance into:

1. Passive Reconnaissance
2. Active Reconnaissance

In passive reconnaissance, you rely on publicly available knowledge. It is the knowledge that you can access from publicly available resources without directly engaging with the target. Think of it like you are looking at target territory from afar without stepping foot on that territory.



Passive reconnaissance activities include many activities, for instance:

- Looking up DNS records of a domain from a public DNS server.
- Checking job ads related to the target website.
- Reading news articles about the target company.

Active reconnaissance, on the other hand, cannot be achieved so discreetly. It requires direct engagement with the target. Think of it like you check the locks on the doors and windows, among other potential entry points.



Examples of active reconnaissance activities include:

- Connecting to one of the company servers such as HTTP, FTP, and SMTP.
- Calling the company in an attempt to get information (social engineering).
- Entering company premises pretending to be a repairman.

Considering the invasive nature of active reconnaissance, one can quickly get into legal trouble unless one obtains proper legal authorization.

Answer the questions below

You visit the Facebook page of the target company, hoping to get some of their employee names. What kind of reconnaissance activity is this? (A for active, P for passive)

P

✔ Correct Answer

You ping the IP address of the company webserver to check if ICMP traffic is blocked. Wh at kind of reconnaissance activity is this? (A for active, P for passive)

A

✔ Correct Answer

You happen to meet the IT administrator of the target company at a party. You try to use social engineering to get more information about their systems and network infrastructure. What kind of reconnaissance activity is this? (A for active, P for passive)

A

✔ Correct Answer

Task 3 Whois

Task 4 nslookup and dig

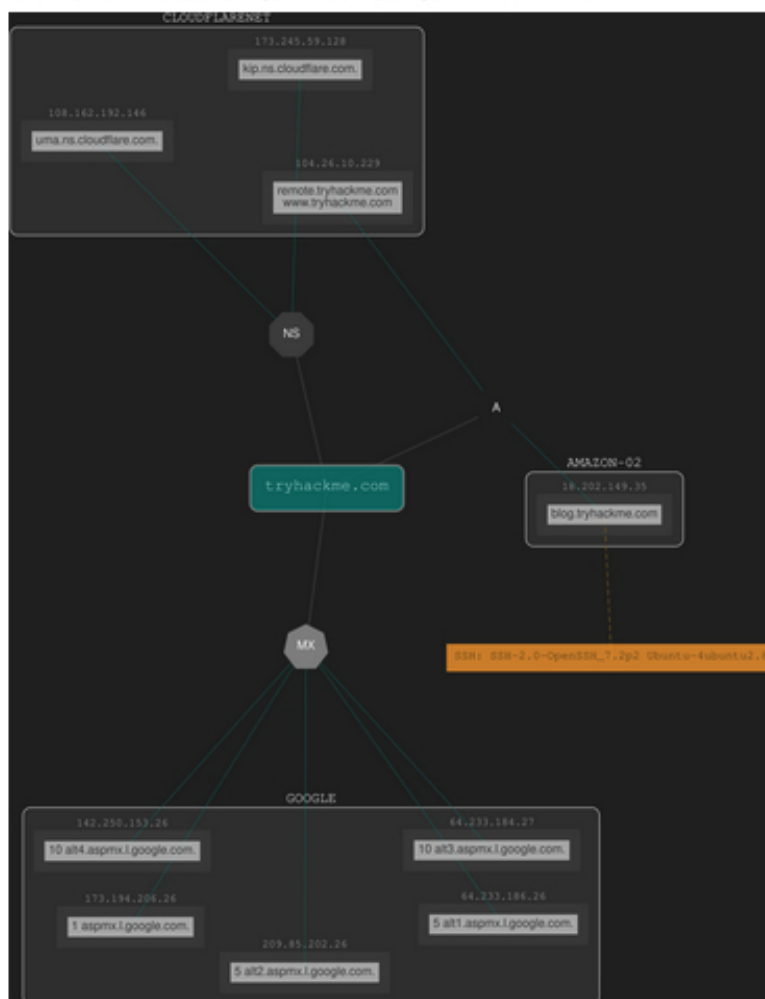
Task 5 DNSDumpster

Task 6 Shodan.io

DNSBurger will also represent the collected information graphically. DNSBurger displayed the data from the table earlier as a graph. You can see the DNS and MX branching to their respective servers, and also showing the IP addresses.



There is currently a beta feature that allows you to export the graph as well. You can manipulate the graph and move blocks around if needed.



Use the web browser on the AttackBox, or your system, to answer the following question.

Answer the questions below

Look up tryhackme.com on twitter.com/tryhackme. What is one interesting outside main that you would discover in addition to `www` and `blog`?

testcode
✓ Correct Answer

Task 8  shodan.io

Task 7 Summary

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Passive Reconnaissance

Learn about the essential tools for passive reconnaissance, such as whois, a lookup, and dig.

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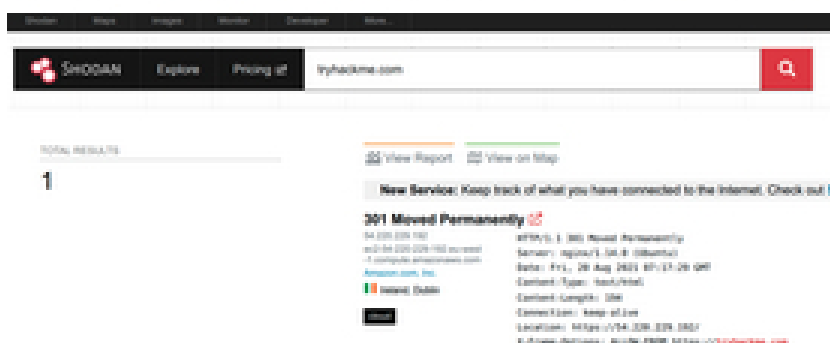
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[Task 1](#) 🟢 Introduction[Task 2](#) 🟢 Passive Versus Active Recon[Task 3](#) 🟢 What is[Task 4](#) 🟢 Whois and dig[Task 5](#) 🟢 DNS Enumeration[Task 6](#) 🟢 Shodan.io

When you are tasked to run a penetration test against specific targets, as part of the passive reconnaissance phase, a service like [shodan.io](#) can be helpful to learn various pieces of information about the client's network, without actively connecting to it. Furthermore, on the defensive side, you can use different services from shodan.io to learn about connected and exposed devices belonging to your organisation.

Shodan.io tries to connect to every device reachable online to build a search engine of connected "things" in contrast with a search engine for web pages. Once it gets a response, it collects all the information related to the service and saves it in the database to make it searchable. Consider the saved record of one of [tryhackme.com](#)'s servers.



This record shows a web server; however, as mentioned already, shodan.io collects information related to any device it can find connected online. Searching for [tryhackme.com](#) on shodan.io will display at least the record shown in the screenshot above. Via this shodan.io search result, we can learn several things related to our search, such as:

- IP address
- hosting company
- geographic location
- server type and version

You may also try searching for the IP addresses you have obtained from DNS lookups. These are, of course, more subject to change. On their [help page](#), you can learn about all the search options available at shodan.io, and you are encouraged to join [TryHackMe's shodan.io](#).

It would be best to visit shodan.io to answer the following questions; however, note that you can find the answers on shodan.io without needing a premium account.

Answer the questions below

According to shodan.io, what is the first country in the world in terms of the number of publicly accessible Apache servers?

✓ Correct Answer

🔍 Hint

Based on shodan.io, what is the third most common part used for Apache?

✓ Correct Answer

🔍 Hint

Based on shodan.io, what is the third most common part used for nginx?

✓ Correct Answer

🔍 Hint

[Task 7](#) 🟢 Summary

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Passive Reconnaissance

Learn about the essential tools for passive reconnaissance, such as `whois`, `nslookup`, and `dig`.

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In this room, we focused on passive reconnaissance. In particular, we covered command-line tools, `whois`, `nslookup`, and `dig`. We also discussed two publicly available services, `dnsdumpster` and `Shodan.io`. The power of such tools is that you can collect information about your targets without directly connecting to them. Moreover, the type of information you may find using such tools can be massive since you master the search options and get used to reading the results.

Purpose	Commandline Example
Lookup WHOIS record	<code>whois tryhackme.com</code>
Lookup DNS A records	<code>nslookup -type=A tryhackme.com</code>
Lookup DNS MX records at DNS server	<code>nslookup -type=MX tryhackme.com 1.1.1.1</code>
Lookup DNS TXT records	<code>nslookup -type=TXT tryhackme.com</code>
Lookup DNS A records	<code>dig tryhackme.com A</code>
Lookup DNS MX records at DNS server	<code>dig 1.1.1.1 tryhackme.com MX</code>
Lookup DNS TXT records	<code>dig tryhackme.com TXT</code>

[Learn more about DNS at DNS in Detail](#)

Answer the questions below

Make sure you note all the points discussed in this room, especially the syntax for the command-line tools.

No answer needed

✓ Correct Answer

How likely are you to recommend this room to others?

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